RAJIV GANDHI COLLEGE OF ENGINEERING AND TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT - 5

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Reg No : 22TD0053

Subject Name : Web Technology

Subject Code : CST63

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Unit : 5

Teacher's Sign :

Department Vision And Mission

Vision

Cultivate student as technocrats, researchers and entrepreneurs in the field of computer science and engineering.

Mission

→ To impart quality education in Computer Science and Engineering by means of learning techniques and value-added courses.

→ To inculcate professional excellence and research culture by encouraging projects in cutting-edge technologies through industry interactions.

→ To build leadership skills, ethical values and teamwork among students.

→ To strengthen the collaboration of department and industry through internships, mentorships and professional body activities.

1) Define SOAP, Mention any security issue of it.

→ SOAP stands for simple object Access protocol.
→ SOAP is communication between application.
→ SOAP is communication protocol.
→ SOAP is a format for sending messages.
→ SOAP communicates via internet

Security issue:

* XML Injection.
* Denial of service
* Authentication and Authorization Issues.
* Fault messages.

2) State the advantages of AJAX.

→ AJAX = Asynchronous Javascript and XML.
→ AJAX is a technique for creating fast and dynamic web pages.

Advantages of AJAX:

1) Performance Enhance.
2) Improved Response Time
3) Reduced server load.
4) Enabled Asynchronous calls.
5) Enhanced user Experience.

3) State the purpose of UDDI.

→ UDDI is a platform-independent framework for describing services, discovering businesses and integrating business services by using the internet.

* UDDI stands for universal description, discovery and integration.
* UDDI is a directory for storing information about web services.
* UDDI communicates via SOAP.

+ UDDI is a directory of web service interfaces described by WSDL.

4) List the benefits of single page web application.

+ Faster Load Times
+ Improved user Experience
+ Reduced server load.
+ Efficient front-end development.
+ Easier Debugging
+ Reusable components.
+ Better caching.

5) State the significance of WSDL document.

→ WSDL stands for web services description language.
→ WSDL is written in XML.
→ WSDL is used to describe web services.
→ WSDL is also used to locate web services.

Significance.

+ Facilitates easy integration of web services into existing system.
+ Interoperability.
+ Tools can automatically generate client code from a WSDL, reducing manual coding and errors

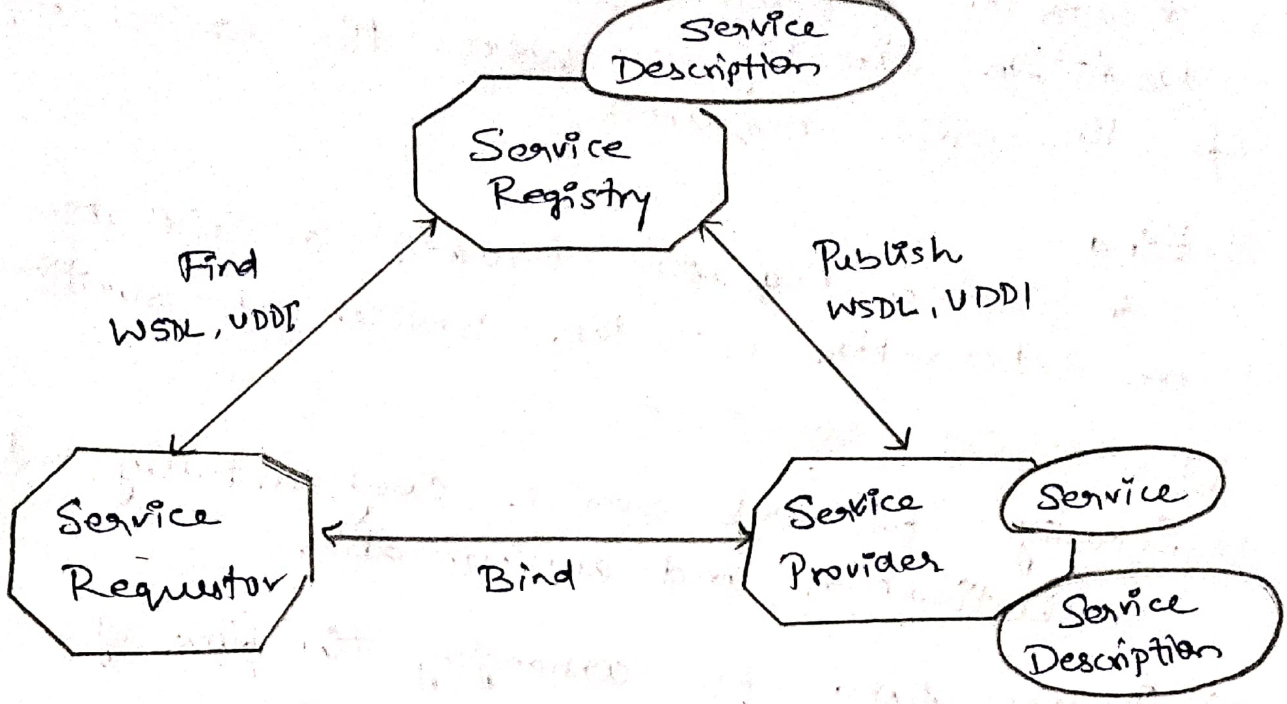
Part B

1) Explain the architecture of web services and how to deploy it?

→ Web services are applications that enable remote procedure calls over a network on the internet often using XML and HTTP.

→ Web service architecture is an interoperability architecture - It identifies those global elements of the global web services network that are required in order to ensure interoperability between web services.

Web service Model:



Roles in web service Architecture:

1) Service Provider
- Owner of the service
- Platform that hosts access to the service.

2) Service Requestor
- Business that requires certain functions to be satisfied.
- Application looking for and invoking an interaction with a service.

3) Service Registry
- Searchable registry of service description where service providers publish their service description.

Operations in web service Architecture:

1) Publish:
- Service descriptions need to be published in order for service requestor to find them.

2) Find :
* Service requestor retrieves a service description directly or queries the service registry for the service required,

3) Bind :
* Service requestor invokes or initiates an interaction with the service at runtime.

Web services specification and technologies.

1) Security - for web services confidentially, integrity, authentication and authorization.

2) Process flow - for arranging the flow of execution across web services.

3) Transactions - for coordinating the results of multiple web services.

4) Messaging - for configuring message paths and routing messages reliably across multiple network hops.

Web services involves three major roles:
1) Service Provider
2) Service Registry
3) Service consumer,

Three major operations surround web services.
* Publishing - making a service available.
* Finding - locating web services.
* Binding - using web services.

Security:
→ Security is one of the most important and most complex issues in the internet and web services.

→ Basic security issues include.

* Data confidentiality and integrity to ensure your credit card number.
* Authentication / authorization, which deals with the rights of individuals or groups to access a certain resource such as given web service interface.

SAML:

→ Security Assertions Markup Language (SAML) provides a standard way to profile information in XML documents and to define user identificatio and authorization information.

XKMS:

The XML Key management specification (XKMS) defines a protocol for distributing and registering public keys used in encrypting and decrypting messages transmitted using SOAP.